MIL-L-52285B
22 December 1975
Superseding
MIL-L-52285A
29 June 1964

MILITARY SPECIFICATION

LIGHT, INSPECTION, FUEL CELL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers a fuel cell inspection light.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

J-C-580	-Cord, Flexible, and Wire, Fixture (Electrical, 0- to 600-Volt Service)
W-C-596	-Connector, Plug, Electrical, Connector, Receptacle, Electrical, Plate, Wall, Electrical
PPP-B-601	Box, Wood, Cleated Plywood

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:12Z
Check the source to verify that this is the current version before use.

PPP-B-636	Box, Shipping, Fiberboard

MILITARY

MIL-P-116	Preservation - Packaging, Methods of
MIL-S-8512	Support Equipment, Aeronautical, Special, General Specification for the Design of
MIL-D-1000	Drawings, Engineering and Associated List

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-143	Specifications and Standards Order of Precedence for Selection of

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL BUREAU OF STANDARDS
Handbook H28 - Screw Thread Standards for Federal Services

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D.C. 20402.)

UNDERWRITERS' LABORATORIES, INC.
Standards for Safety (UL844 - Electric Lighting Fixtures for Use in Hazardous Locations)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:12Z
Check the source to verify that this is the current version before use.

(Application for copies should be addressed to the Underwriters' Laboratories, Inc., 207 East Ohio Street, Chicago, IL 60611; or 333 Pfingston Road, Northbrook, IL 60062; or 2602 Tampa East Blvd., Tampa, FL 33619; or 1655 Scott Boulevard, Santa Clara, CA 95050; or 1285 Walt Whitman Road, Melville, L.I., NY 11746.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT:
Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Design. The inspection light shall be a light and mirror assembly adequate for inspecting aircraft fuel cells and inaccessible areas. It shall be light weight and safe for use in hazardous locations. Proof shall be submitted that it has the approval of the Underwriters' Laboratories, Inc. as an electric light fixture suitable for use in class I, group D hazardous locations. Listing by the Laboratories will be accepted as evidence of approval. Detailed requirements not specified herein shall conform to the applicable provisions of Specification MIL-S-8512 for aircraft support equipment.

3.2 Materials. Materials used in the construction of the light shall be nonsparking. Otherwise, materials shall be as specified herein and as specified in Specification MIL-S-8512. When specifications and standards are not specifically designated, selection of materials and processes shall be in accordance with MIL-STD-143.

3.3 Components. General arrangement of components shall be as shown on figure 1.

3.3.1 Subassemblies. The subassemblies shall consist of a forward and a rear section.

3.3.1.1 Forward section. The forward section shall consist of two subsections joined by a threaded collar. The subsections shall be capable of being position-locked to a collar every one-quarter turn by a recessed lockscrew. The forward section shall be explosion-proof and shall have a 40-watt, 115- to 125-volt, standard T-6½ shape, rough-service lamp enclosed within an explosion-proof, heat-resistant globe located at the mirror end. The forward section shall have a 7/8-inch

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:12Z
Check the source to verify that this is the current version before use.

hole, + 1/8-inch, cut away on the end to transmit the lamp beam, and three equidistant slots cut away on the side, with one slot having a longitudinal centerline in the same plane as the forward section trajectory (see fig. 2), allowing at least 60 percent of the lamp beam to be transmitted. Position locking of the threaded collar shall allow for firm cushion mounting of the lamp and enclosing globe, and for replacement of the lamp without the use of special tools.

3.3.1.2 Rear section. The rear section shall house the tubular lamp wiring. It shall act as a handgrip and extension. One end shall connect to the forward section, and the other end shall provide for an explosion-proof device for attaching the electric cord assembly (see 3.3.3).

3.3.2 Mirror assemblies. Three interchangeable mirror assemblies, each consisting of a mirror, bracket, and attaching device for mounting and manual control (see fig. 1), shall be provided. Attaching device shall be connected to shortest side of the mirror. The mirrors shall be of fuel-fume-resistant specular glass having a minimum of 85 percent reflection. Their surfaces shall be positioned flush with the edges of the mounting device, to prevent edge shadows from the mounting device being reflected.

3.3.3 Electric cord assembly. An electric cord assembly shall connect to the explosion-proof device on the rear section (see 3.3.1.2), and extend 75 feet to a two-pole, three-wire, polarized, 15-ampere plug conforming to style D23 of Specification W-C-596.

3.3.3.1 Electric cord. The electric cord for the cord assembly shall be #18 AWG three-conductor cable in conformance with type V or SJO of Specification J-C-580.

3.3.4 Controls. Manually operated control rods shall be provided for manual positioning of the mirror assembly and forward section (see 3.5.1). The force required to operate the controls shall be in accordance with the force limitations specified in Specification MIL-S-8512.

3.3.5 Latch assembly. A latch assembly shall be attached to the rear section to support and guide the control rods and provide for position locking of the forward section control (see 3.3.4). The assembly design shall allow the operator to position lock and unlock the forward section control rod with the hand holding the light, for the other hand to be free for manual positioning of the control rods.

3.4 Construction. The inspection light shall be constructed to withstand without damage all normal conditions of service use without loosening

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:12Z
Check the source to verify that this is the current version before use.

of parts, permanent deformation, or loss of serviceability (see 3.2).

3.4.1 Dimensions. Inspection light dimensions shall be as shown on figures 1 and 2.

3.4.2 Weight. Total weight of the light assembly shall not exceed 7-1/2 pounds.

3.4.3 Part numbering of interchangeable parts. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of Specification MIL-D-1000 shall govern the manufacturer's part numbers and changes thereto.

3.4.4 Threaded parts. Threaded parts shall conform to Handbook H28. The American National fine thread series shall be used for threaded parts less than 1/4-inch diameter. The Unified or American National coarse thread series shall be used for threaded parts 1/4-inch diameter and larger; however, the American National fine thread series may be employed for these sizes when applicable. Pipe threads shall conform to the American National taper pipe thread series.

3.5 Performance.

3.5.1 Mirror assemblies. The mirror assemblies shall be capable of being independently positioned (see 3.3.4) from a parallel plane with the forward section, through a 75-degree fixed trajectory, $\pm$ 5 degrees (see fig. 2).

3.5.2 Forward section. The forward section shall be capable of being independently positioned from a horizontal plane with the rear section, through a 60-degree fixed trajectory, $\pm$ degrees (see fig. 2).

3.6 Finish. Finish of the inspection light shall be as specified in Specification MIL-S-8512.

3.7 Identification. Nameplate and markings shall be as specified in Specification MIL-S-8512.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:12Z
Check the source to verify that this is the current version before use.

3.7.1 Caution plate. A plate reading as follows shall be attached to the assembly.

CAUTION

INSURE THAT THE LIGHT AND AIRCRAFT ARE GROUNDED TO A COMMON GROUND BEFORE PLACING THE LIGHT IN THE AIRCRAFT FLUID TANK.
THIS LIGHT TO BE SERVICED (INCLUDING RELAMPING) ONLY BY QUALIFIED PERSONNEL AND ONLY IN APPROVED AREAS WELL REMOVED FROM FUEL SYSTEM MAINTENANCE AREAS.

3.7.2 Labeling. Light shall be labeled in accordance with Standard UL844.

3.8 Instruction data. If specified (see 6.2), complete operating and maintenance instructions, and a repair parts listing shall be furnished with each light. Two copies shall be furnished the contracting officer.

3.9 Workmanship. The inspection light, including all parts and accessories, shall be a unit free of all defects which could affect proper functioning in service. Workmanship shall be in accordance with that specified in Specification MIL-S-8512 for aeronautical support equipment.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the government. The government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Sampling for lot acceptance.

4.2.1 Lot. A lot shall consist of all light assemblies submitted for inspection at one time.

4.2.2 Examination. Examination shall be in accordance with Standard MIL-STD-105. Samples shall be examined for defects listed in Table I, with the acceptable quality levels (AQL's) as specified.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:12Z
Check the source to verify that this is the current version before use.

TABLE I. Categories of defects in accordance with Standard MIL-STD-105.

Classification	AQL percent defective	Item	Paragraph reference	Method of inspection
CRITICAL	None	Nonsparking material has not been used.	3.2	Visual
MAJOR:	2.5			
101	.....	Cushion mounting of lamp and enclosing globe not firm. Sixty percent of lamp not exposed.	3.3.1.1	Visual
102	.....	Mirror is not positioned flush with its mounting device.	3.3.2	Visual
103	.....	Part or component missing, including caution plates and labels.	3.7	Visual
104	.....	Mating components do not fit...	3.9	Manual
105	.....	Weight not as specified.....	3.4.2	Manual
106	.....	Material not as specified.....	3.2	Visual
MINOR:	6.5			
201	.....	Dimensions not as specified....	3.4.1	Visual
202	.....	Finish not as specified.....	3.6	Visual
203	.....	Improper identification marking	3.7	Visual
204	.....	Burrs and rough edges not removed from machined surfaces.	3.9	Visual and manual

Source: https://assist.dla.mil -- Downloaded: 2026-02-22T21:12Z
Check the source to verify that this is the current version before use.

4.3 Tests. Each inspection light shall be completely assembled and connected to an adequate power supply to determine if proper electrical connections have been accomplished. The light assembly shall then be operated to determine force limitations (see 3.3.4), latch assembly locking (see 3.3.5), and performance (see 3.5).

4.4 Preparation for delivery. Preservation, packaging, packing, and marking shall be inspected to assure compliance with section 5 of this specification.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A, B, or C, as specified (see 6.2).

5.1.1 Level A. The mirrors of the assembly shall be removed and packaged together in accordance with 1A-8 of MIL-P-116. Each mirror shall be cushioned to prevent damage. The lamp shall be removed and packaged in accordance with method III of MIL-P-116. The remainder of the assembly shall be packaged in accordance with method III of MIL-P-116. The three items comprising the assembly shall then be packaged together in a fiberboard box equal to W6s or W6c of PPP-B-636. Items shall be cushioned to prevent movement within the container.

5.1.2 Level B. Packaging shall be as for Level A except that the unit container may conform to Domestic Class of PPP-B-636.

5.1.3 Level C. Preservation and packaging shall be sufficient to afford protection against damage during shipment from the supply source to the first receiving activity.

Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A. Light assemblies packaged as specified in 5.1 shall be packed in wood-cleated plywood boxes conforming to Grade A of PPP-B-601. Gross weight of each box shall not exceed 200 pounds.

5.2.2 Level B. Light assemblies packaged as specified in 5.1 shall be packed in snug-fitting, fiberboard boxes equal to V3s, V4s or V3c of PPP-B-636.

5.2.3 Level C. The assemblies shall be packed in a manner to insure carrier acceptance and safe delivery to destination at the lowest applicable rate. Shipping containers shall comply with the Uniform Freight

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:12Z
Check the source to verify that this is the current version before use.

Classification Rules or with other carrier regulations applicable to the mode of transportation.

5.3 Marking. In addition to any special marking required by the contract or order, marking shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The inspection light covered by this specification is for use in inspecting inaccessible aircraft fuel cell areas where explosive gases may be present.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Whether instruction data are required (see 3.8 and 6.3).

(c) Level of preservation and packaging, and level of packing required (see 5.1 and 5.2).

(d) Whether special marking is required (see 5.3).

6.3 Contract data requirements. Data specified in paragraph 3.8, if required, shall not be delivered to the procuring activity unless specified on the Contract Data Requirements List (DD 1423) referencing the appropriate data item description in the military departments Authorized Data List (ADL).

Custodians:

Army - AV
Navy - AS
Air Force - 82

Review Activity:

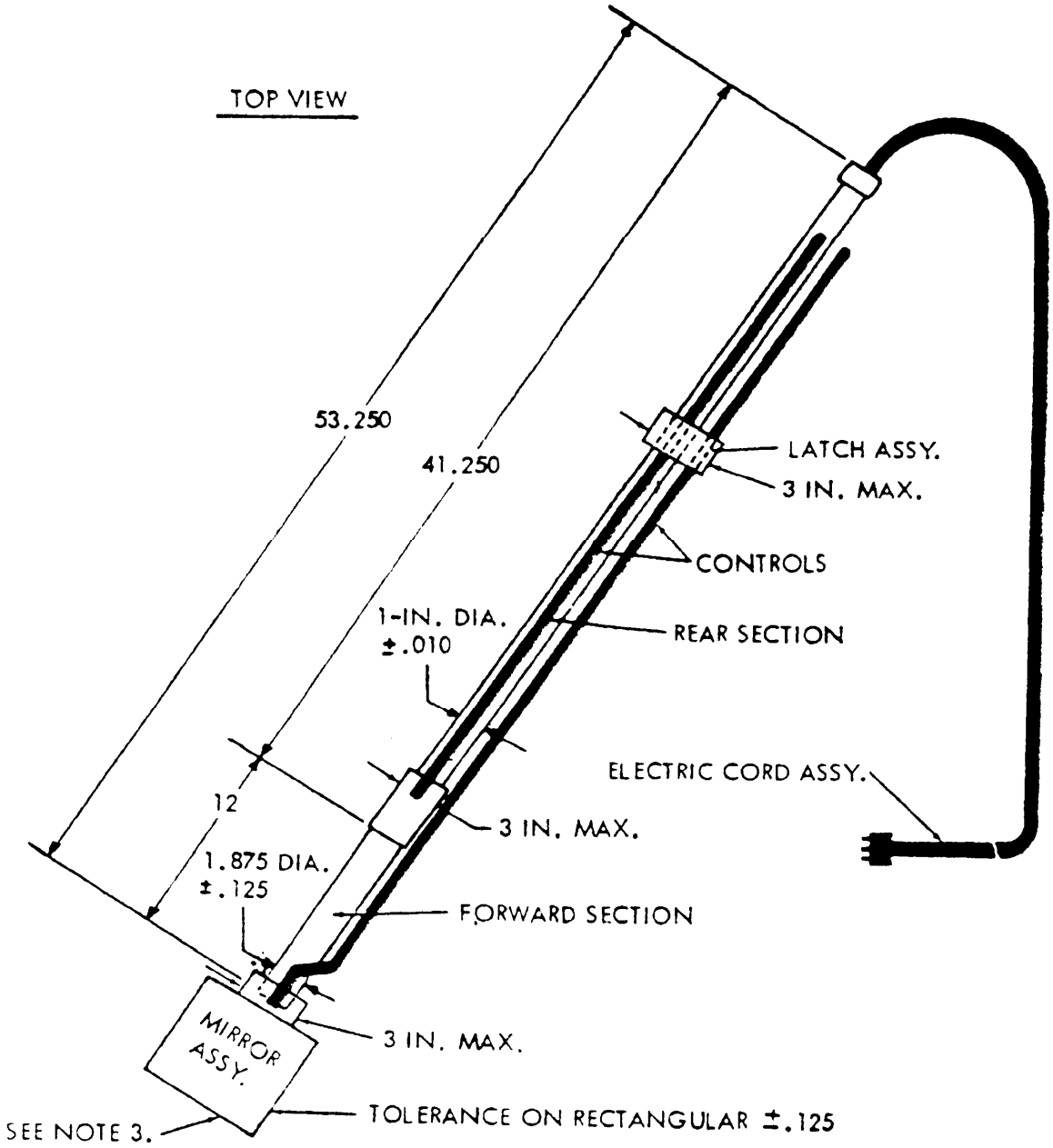
DSA-GS

Preparing Activity:

Army - AV

Project No. 6230-0223

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:12Z
Check the source to verify that this is the current version before use.



NOTES:
1. ALL DIMENSIONS ARE IN INCHES.
2. UNLESS OTHERWISE SPECIFIED, TOLERANCES ON DIMENSIONS ARE ±.250
3. MIRROR ASSEMBLIES FURNISHED SHALL HAVE RECTANGULAR DIMENSIONS OF 3 x 6, 3 x 4, 4 x 4 INCHES.

FIGURE 1. INSPECTION LIGHT DIMENSIONS AND GENERAL ARRANGEMENT

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:12Z
Check the source to verify that this is the current version before use.

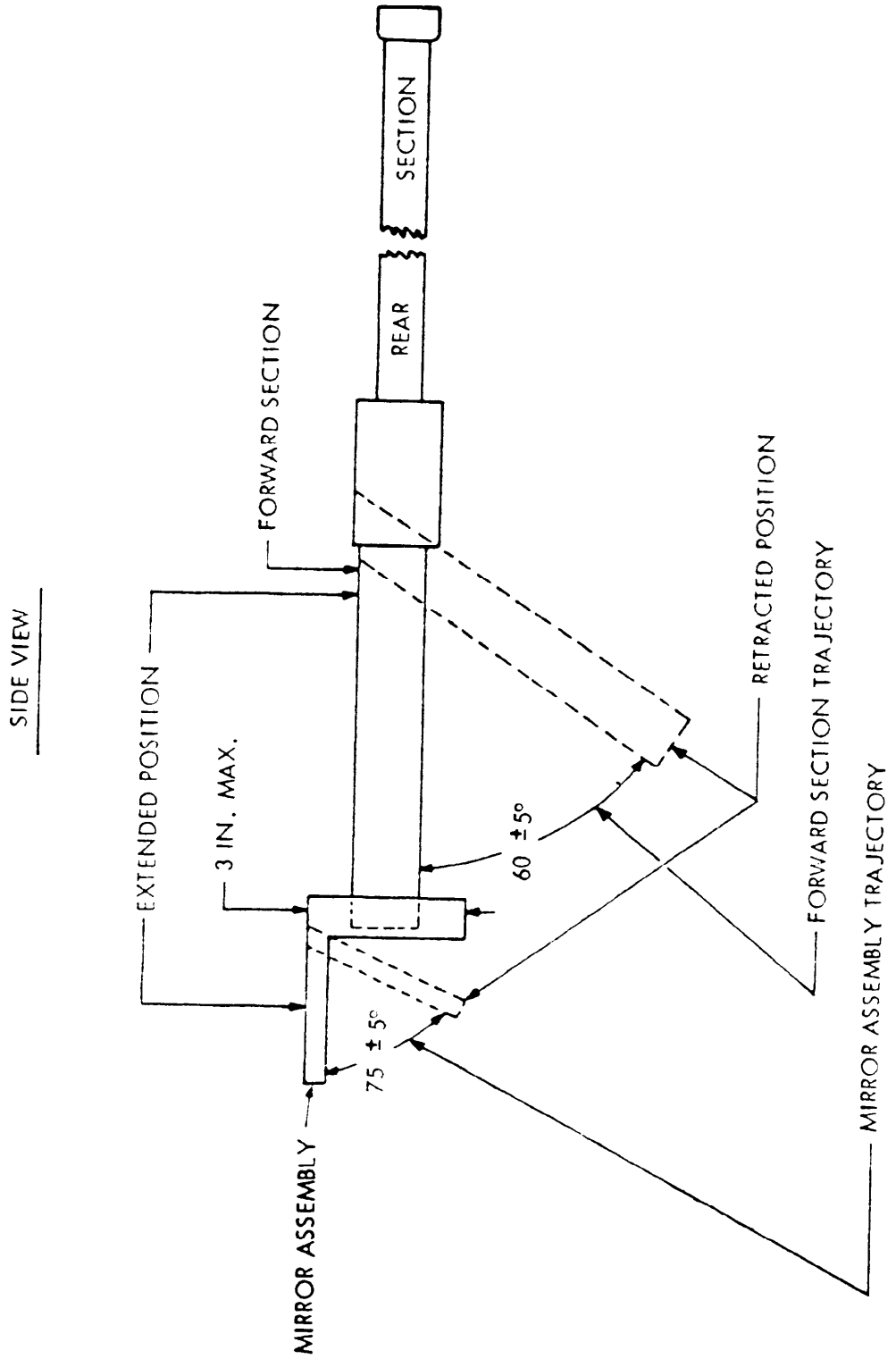


FIGURE 2. POSITIONING OF MIRROR ASSEMBLY AND FORWARD SECTION, FUEL CELL INSPECTION LIGHT

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:12Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/ vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4 REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426

REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

S/N 0102-014-1802

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:12Z
Check the source to verify that this is the current version before use.

FOLD

POSTAGE AND FEES PAID



OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

COMMANDER
U.S. ARMY AVIATION SYSTEM COMMAND
ATTN: AMSAV-EKS
P.O. BOX 209 MAIN OFFICE
ST.LOUIS, MO. 63156

FOLD

U.S. GOVERNMENT PRINTING OFFICE 1976—

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:12Z
Check the source to verify that this is the current version before use.